# LLM 기반 네트워크 프로토콜 명세서의 보안 요구사항 추출: 동향 조사

Hassan Asghar*, 이현우†

*,† 한국에너지대학교 (대학원생, 교수)

# LLM-Based Security Requirement Extraction from Network Protocol Specifications: A Survey

Hassan Asghar*, Hyunwoo Lee†

*,† Korea Institute of Energy Technology (Graduate student, Faculty)

**요 약**

Large language models (LLMs) are increasingly used to analyze network protocol specifications and extract security-relevant requirements. This survey focuses on the path from specification text to formal security property, distinguishing requirement extraction from property formalization. We review six studies covering security-requirement extraction, requirement formalization, and LLM-assisted verification. We compare their inputs, outputs, security focus, and evaluation methods. Three challenges recur: similarity-based metrics may miss security-critical changes in quantifiers, session bindings, temporal conditions, and negation; coverage is difficult to compare across different annotation units and denominators; and verification success does not establish that an extracted property faithfully represents the source specification.. We identify four evaluation criteria: source traceability, semantic correctness, coverage, and verification evidence.

## I. Introduction

Network protocol specifications are long documents that combine message formats, state behavior, normative requirements, exceptions, and security considerations. Security requirements may be distributed across multiple sections and expressed in different forms. This makes automatic identification of security-relevant requirements from protocol specifications a challenging language-understanding task [1].

Recent LLM-based methods address this problem by identifying security-relevant statements and, in some cases, formalizing them as machine-checkable security properties. These are distinct steps: extraction determines whether the relevant security requirement is identified from the specification, while formalization determines whether that requirement is expressed correctly in a formal language. This survey focuses on this specification-to-property process; protocol modeling and verification are considered as downstream context rather than the primary focus.

This survey addresses three research questions: **(RQ1)** How do LLM-based methods extract security requirements from network protocol specifications and formalize them as security properties? (**RQ2**) How are extraction and formalization quality evaluated? (**RQ3**) What evidence is provided that the extracted or formalized property preserves the security meaning of the source specification?

We review six representative studies published between 2022 and 2026. Studies were selected based on their primary contribution to security-requirement extraction [2][3], formalization of security requirements into formal properties [4][5], or LLM-assisted generation and analysis of formal security properties [6][7]. Studies focused primarily on protocol-model construction were excluded. We treat this as a focused survey rather than an exhaustive systematic review and compare the evaluation evidence provided by the included studies. Figure 1 illustrates the survey scope from protocol specification to security requirement and formal security property, with LLM-assisted verification shown as a downstream use of the extracted property.

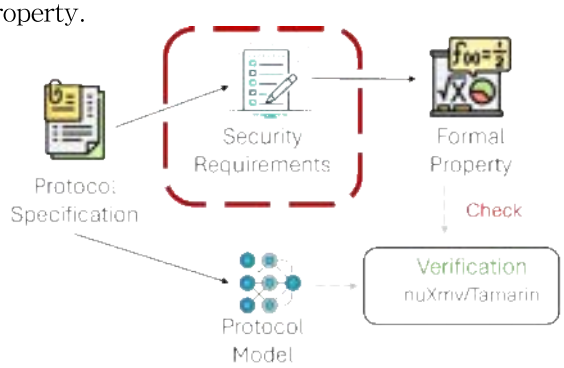



**Fig. 1**. Scope of the survey: LLM-based extraction and formalization of security requirements from network protocol specifications, with downstream formal analysis shown for context.

## II. Terminology and Classification

**A.Scope & Classification.** The reviewed literature is classified into three areas: security-requirement extraction, requirement formalization, and LLM-assisted verification or formal-property generation. Security-requirement extraction identifies security-relevant statements in a specification. Requirement formalization expresses a selected requirement as a machine-checkable property. LLM-assisted verification refers to methods that use LLMs with formal tools to generate, analyze, or validate formal security properties. Protocol-model construction is outside the scope of this survey. A single study may address multiple tasks, but the tasks should be evaluated separately. Table 1 summarizes the three in-scope categories; protocol-model construction is excluded because it is not the primary focus of this survey.

| | |
|---|---|
| Security-requirement extraction [2][3] | Identify security-relevant statements or goals from protocol documents. |
| Requirement formalization [4][5] | Translate selected natural-language requirements into formal security properties. |
| LLM-assisted verification [6][7] | Generate, analyze, or validate formal security properties using LLMs and formal verification tools. |

**Table 1**. Categorization of the Reviewed Studies.

**B.Source Traceability**. A formal property should be linked to the specification clauses from which it is derived. If a requirement depends on multiple clauses or assumptions, the supporting text should be identified explicitly.

**C.Semantic Correctness**. The formal property should preserve the security meaning of the requirement, including principals, resources, session bindings, freshness conditions, quantifiers, ordering constraints, exceptions, and negations. Syntactic validity alone does not establish semantic correctness.

**D.Example**. A specification may require a server to accept a request only after authenticating the client in the same session. A formal property that permits authentication from a different session may be textually similar but express a weaker security condition. This illustrates why

security-property evaluation should examine semantic correctness rather than textual similarity alone.

## III. Literature Review

SPARTA and SecGoal directly address security-requirement or security-goal extraction from protocol documents [2], [3]. SPARTA uses retrieval-augmented processing to identify and structure security-relevant clauses, while SecGoal provides expert-annotated security goals from protocol documents [2], [3]. These studies provide evidence for extraction performance but do not demonstrate complete specification-to-verification pipelines.

nl2spec translates natural-language requirements into temporal logic with interactive correction [4]. Li et al. investigate document-to-formal-specification extraction for test automation rather than security properties specifically [5]. These studies are relevant to property formalization but do not directly evaluate security-requirement extraction.

CryptoFormalEval, and PropertyGPT use LLMs with formal tools such as ProVerif or Tamarin to generate, analyze, or validate formal properties [6], [7]. These related methods provide evidence about downstream formal-property generation and analysis rather than direct extraction from network protocol specifications. Their verification results therefore do not establish that security requirements were correctly extracted from the original specification.

## IV. Evaluation Challenges

**1) Similarity metrics do not establish semantic correctness.** A formal property is defined by its logical meaning, not by textual similarity. For example, replacing a same-session condition with an existentially quantified session may produce a similar-looking property while weakening the security guarantee. Similarity metrics can therefore be diagnostic, but they cannot replace semantic evaluation.

**2) Coverage depends on the annotation unit.** Studies may report coverage over sentences, clauses, security goals, properties, or state transitions. Results are not directly comparable unless the unit, denominator, inclusion criteria, and treatment of omissions are reported.

**3) Verification does not establish specification correctness.** A verifier checks a formal model against a formal property, not against the original specification. If the model and property omit the same condition, verification may still succeed. Therefore, correctness of an extracted property requires independent evidence linking it to the source specification.

## V. Research Directions

**Specification-Grounded Benchmarks.** Future benchmarks should link each security requirement to its supporting specification clauses and include contrastive pairs that differ in one security-critical detail, such as a principal, session binding, freshness condition, or quantifier. Such benchmarks can test semantic correctness beyond surface similarity. The proposed evaluation criteria are intended to be protocol-independent, while their concrete application should preserve protocol-specific security semantics. Thus, the evaluation methodology can be shared across protocol families, but the security requirements and formal properties used for evaluation should reflect protocol-specific mechanisms and terminology.

**End-to-End Evaluation**. Studies should evaluate extraction, formalization, and verification separately before reporting end-to-end results. They should also report model versions, experimental settings, expert corrections, and rejected outputs to make reproducibility and human effort visible.

## VI. Conclusion

LLM-based security-property extraction from network protocol specifications involves identifying security requirements and formalizing them as machine-checkable properties, with LLM-assisted verification providing downstream evidence. The reviewed studies show that extraction and formalization require separate evaluation, particularly with respect to source grounding and semantic correctness. We therefore recommend evaluating source traceability, semantic correctness, coverage, and verification evidence separately.

## Acknowledgement

This work was supported by Institute of Information & communications Technology Planning & Evaluation (IITP) under the Artificial Intelligence Innovation Human Resources Development (IITP-2026-RS-2026-25548323) grant funded by the Korea government (MSIT).

## [References]

[1] I. Karim, H. Lee, H. Asghar, K. S. Mubasshir, S. Han, M. H. Bhuiyan, and E. Bertino, “VWAttacker: A Systematic Security Testing Framework for Voice over WiFi User Equipments,” IEEE International Conference on Computer Communications (INFOCOM), 2026, doi:10.1109/INFOCOM59046.2026.11571377.

[2] H. Asghar et al., “Poster: Automated Security Property Extraction from Protocol Specifications,” 2025 IEEE International Conference on Network Protocols (ICNP), 2025, doi:10.1109/ICNP65844.2025.11192438.

[3] D. Huang et al., “SecGoal: A Benchmark for Security Goal Extraction and Formalization from Protocol Documents,” arXiv:2604.27601v1, 2026.

[4] M. Cosler et al., “nl2spec: Interactively Translating Unstructured Natural Language to Temporal Logics with Large Language Models,” CAV 2023, LNCS 13965, pp. 383 - 396, 2023, doi:10.1007/978-3-031-37703-7_18.

[5] H. Li et al., “Extracting Formal Specifications from Documents Using LLMs for Test Automation,” IEEE/ACM ICPC 2025, pp. 310 - 321, 2025.

[6] C. Curaba, D. D’Ambrosi, A. Minisini, and N. Pérez-Campanero Antolín, “CryptoFormalEval: Integrating LLMs and Formal Verification for Automated Cryptographic Protocol Vulnerability Detection,” arXiv:2411.13627, 2024.

[7] Y. Liu et al., “PropertyGPT: LLM-driven Formal Verification of Smart Contracts through Retrieval-Augmented Property Generation,” NDSS Symposium 2025, 2025, doi:10.14722/ndss.2025.241357.